

Agricultural Demand Prediction & Decision Support System

Comprehensive Technical Architecture, Languages, Databases & Machine Learning Algorithms

1. Executive Summary & Project Overview

The **Hanu Agri Demand Prediction System** is an end-to-end intelligent agricultural advisory platform designed to empower Indian farmers, agronomists, and supply-chain stakeholders. The system combines multi-variable supervised machine learning models, statistical time-series forecasting, agro-climatic formulas, and geospatial analytics to deliver real-time actionable insights for crop recommendation, commodity price trends, fertilizer schedules, market glut risk, precision irrigation, and crop disease diagnosis.

2. Frontend Technologies & Languages

Layer / Component	Technology & Version	Implementation Details & Capabilities
Markup	HTML5	Semantic structure for 10+ modules, responsive dashboards, modal overlays, input forms, and data cards.
Styling	CSS3 (Modern Vanilla)	Custom design system, CSS Grid/Flexbox, glassmorphism, responsive breakpoints, smooth micro-interactions, dark accents.
Client Logic	JavaScript (ES6+)	Asynchronous REST client (Fetch API), dynamic DOM updates, state management, client-side routing, voice TTS/STT handling.
Charts & Visuals	Chart.js (v4.4.0)	Interactive 30-day price trend lines, uncertainty intervals, ROI donut charts, and weather precipitation graphs.
Geospatial Maps	Leaflet.js (v1.9.4) & OpenStreetMap	Interactive APMC Mandi geolocation mapping, custom markers, user coordinate radius, and route discovery.
Voice & Speech	Web Speech API	Integrated voice-driven chatbot queries and multilingual text-to-speech audio playback in English, Kannada, and Hindi.

3. Backend Technologies & Frameworks

Component	Technology	Architectural Role
Programming Language	Python 3.11+	Core runtime for data processing pipelines, ML inference, mathematical computations, and REST services.
Web Microframework	Flask (v3.0+)	Lightweight, high-throughput REST API server handling request validation, routing, JSON serialization, and static asset delivery.
Cross-Origin Handler	Flask-CORS (v4.0+)	Manages secure cross-origin resource access across web clients, endpoints, and external callers.
Scientific ML Stack	Scikit-Learn, NumPy, Pandas	High-performance matrix computing, feature scaling, model fitting, and time-series feature engineering.
Image Engine	Pillow (PIL v10.0+)	Decodes, resizes, and pre-processes leaf image buffers for plant pathology detection.

4. Database & Storage Architecture

The system uses a resilient, high-speed **serialized & file-based storage architecture** designed for low latency and zero-dependency cloud deployment:

Storage Medium	File Path / Asset	Contents & Functionality
Tabular CSV Datasets	backend/data/*.csv	Contains ground-truth crop parameter datasets (crop_data.csv), fertilizer records (fertilizer_data.csv), and historical mandi modal prices across 10+ states (price_data.csv).
JSON Knowledge Bases	backend/data/*.json	Relational structured knowledge stores for Indian government agricultural subsidies (PM-KISAN, PMFBY, PKVY) and plant pathology treatment guides (disease_labels.json).
Pre-Trained Pickles	backend/models/*.pkl	Pre-compiled binary Scikit-Learn pipelines containing trained decision trees, ensemble estimators, scalers, and label encoders for instant <10ms inference.

5. Comprehensive Mathematical & Machine Learning Algorithms

The platform integrates eight core machine learning models, statistical algorithms, and domain-specific agro-climatic formulas:

A. Crop Recommendation — Random Forest Classifier

Model: RandomForestClassifier(n_estimators=200, max_depth=20, min_samples_split=5)

Inputs: Nitrogen (N), Phosphorus (P), Potassium (K), Temperature (°C), Humidity (%), Soil pH, Rainfall (mm).

Pre-processing: Z-score standard normalization:

$$z = (x - \mu) / \sigma$$

Mechanism: Constructs an ensemble of 200 de-correlated decision trees trained via bootstrap aggregating (bagging). Outputs calibrated class probabilities for 22+ agricultural crops, returning the top 5 ranking crops with percentage confidence scores.

B. Price & Demand Forecasting — Gradient Boosting Regressor

Model: GradientBoostingRegressor(n_estimators=200, max_depth=6, learning_rate=0.1)

Time-Series Feature Engineering: To capture seasonal cycles and harvest periodicity, dates are converted to harmonic circular features:

$$\text{month_sin} = \sin(2\pi \times \text{month} / 12), \text{month_cos} = \cos(2\pi \times \text{month} / 12)$$

$$\text{doy_sin} = \sin(2\pi \times \text{doy} / 365), \text{doy_cos} = \cos(2\pi \times \text{doy} / 365)$$

Mechanism: Sequentially trains weak regression trees minimizing Mean Squared Error (MSE). Generates 30-day projected commodity prices along with asymmetric confidence bands (92% min floor, 108% max ceiling) and directional momentum analysis.

C. Fertilizer Recommendation — Multi-Class Decision Tree

Model: DecisionTreeClassifier(max_depth=15, min_samples_split=5)

Inputs: Soil Type, Crop Type, Nitrogen, Phosphorus, Potassium, Moisture, Temperature, Humidity.

Mechanism: Employs Gini Impurity recursive partitioning to recommend the exact fertilizer formula (Urea, DAP, MOP, NPK 20-20-20, SSP, etc.) supplemented with automated agronomic nutrient threshold warnings (Low/High/Optimal).

D. Market Glut & Overproduction Risk Engine

Algorithm: Demand-Supply Elasticity & Risk Heuristic Algorithm.

Mathematical Formulation:

$$\text{Supply-Demand Ratio (R)} = \text{Projected Production (MT)} / \text{Estimated Demand (MT)}$$

$$\text{Risk Score} = \text{clamp}(10, 95, (R - 0.85) \times 200)$$

Output: Classifies overproduction risk into Low (<40), Moderate (40-65), or High (>65), warning farmers against catastrophic price crashes and suggesting lower-risk alternative diversification crops.

E. Geospatial Mandi Proximity — Haversine Distance Formula

Algorithm: Great-Circle Geodesic Distance Computation.

Mathematical Formulation:

$$a = \sin^2(\Delta\phi / 2) + \cos(\phi_1) \times \cos(\phi_2) \times \sin^2(\Delta\lambda / 2)$$

$$d = 2 \times R \times \arcsin(\sqrt{a}), \text{ where } R = 6,371 \text{ km}$$

Output: Computes exact geodesic distance between farmer device coordinates and regional wholesale APMC mandis.

F. Evapotranspiration & Precision Irrigation Advisory

Algorithm: Hargreaves-Samani Evapotranspiration Formula (ET_0).

Mathematical Formulation:

$$ET_0 = 0.0023 \times (T_{\text{mean}} + 17.8) \times (T_{\text{max}} - T_{\text{min}})^{0.5} \times R_a$$
$$\text{Crop Water Requirement (Liters/Acre)} = ET_0 \times 4,046.86$$

Output: Precise daily irrigation volume recommendation, rainfall wash-off spraying alerts, and fungal pest humidity risk indexing.

G. Crop ROI & Farm Budgeting Matrix

Mathematical Model:

$$\text{Total Cost} = \Sigma(\text{Seeds} + \text{Land Prep} + \text{Fertilizers} + \text{Labor} + \text{Irrigation} + \text{Harvest}) \times \text{Acres}$$

$$\text{Gross Revenue} = (\text{Yield per Acre} \times \text{Acres}) \times \text{Market Price per Quintal}$$

$$\text{Net Profit} = \text{Gross Revenue} - \text{Total Cost}, \text{ ROI \%} = (\text{Net Profit} / \text{Total Cost}) \times 100$$

Output: Full financial feasibility breakdown and automated ROI comparison ranking against substitute crops.

H. Plant Leaf Disease Pathology Detection

Architecture: Convolutional Neural Network (CNN) Deep Learning Classifier pipeline trained on agricultural pathology datasets (PlantVillage). Evaluates leaf lesion patterns to output disease diagnoses, severity levels (Low/Medium/High), and biological/chemical remedy protocols.

6. External Cloud Services & APIs

- **Open-Meteo Weather & Geocoding API:** Provides free, high-resolution live meteorological data and coordinate lookup.
- **WeatherAPI.com:** Enterprise fallback live weather forecasting provider.
- **OpenStreetMap Nominatim:** Open-access reverse geocoding and tile server mapping for APMC mandi navigation.
- **HTML5 Web Speech API:** Multilingual speech recognition and acoustic synthesis for vernacular accessibility.